

# Benchmark results compiled with -O0: mean time (microseconds)

| Test case               | heftia    | freer          | polysemy  | fused     | effectful      | eff       | effective       | effective.naive | mp       | mp.safe  | mtl.logict | handrolled.logict | effective.lstg | mtl.listt | handrolled.listt |
|-------------------------|-----------|----------------|-----------|-----------|----------------|-----------|-----------------|-----------------|----------|----------|------------|-------------------|----------------|-----------|------------------|
| countdown.shallow.10000 | 798.538   | <b>509.13</b>  | 1733.018  | 1749.357  | 740.727        | 896.962   | 1614.533        | 1625.73         | 845.102  | 2174.377 | 654.79     | 565.603           | –              | –         | –                |
| countdown.deep.10000    | 3004.258  | 978.679        | 2890.541  | 10750.331 | <b>744.925</b> | 885.597   | 1677.138        | 3247.911        | 1447.842 | 5357.541 | 5238.784   | 5922.169          | 1689.268       | –         | –                |
| catch.shallow.10000     | 1083.237  | 310.476        | 2808.698  | 3467.633  | 2154.685       | 379758.1  | 963.012         | 1090.171        | 1661.942 | 1658.47  | 256.513    | <b>226.925</b>    | –              | –         | –                |
| catch.deep.10000        | 8351.972  | <b>317.406</b> | 7077.556  | 16239.5   | 2193.773       | 412260    | 1033.715        | 2981.111        | 1670.427 | 1655.556 | 3962.116   | 4114.517          | 1119.016       | –         | –                |
| local.shallow.10000     | 4518.962  | 674.51         | 2651.992  | 1430.328  | 1072.911       | 416023    | 629.987         | 666.496         | 1406.982 | 1399.893 | 352.422    | <b>329.673</b>    | –              | –         | –                |
| local.deep.10000        | 13442.725 | 678.417        | 5617.95   | 5973.106  | 1082.439       | 394305.5  | <b>636.916</b>  | 1640.293        | 1414.94  | 1409.345 | 818.892    | 1608.341          | 651.544        | –         | –                |
| nondet.shallow.32       | 7761.631  | 6054.539       | 13238.856 | 2138.505  | –              | 28539.737 | 4167.837        | 4231.028        | 3094.095 | 3069.472 | 2469.588   | <b>1978.713</b>   | –              | 7458.944  | 7026.009         |
| nondet.deep.32          | 15788.212 | 8541.178       | 22926.463 | 10690.034 | –              | 54767.725 | <b>4287.117</b> | 9690.469        | 7770.347 | 7847.113 | 6804.812   | 7173.219          | 4540.744       | 80644.95  | 80662.85         |